MODULE 1 · LECTURE 7

Introduction to the Internet and WWW: Concepts, Components and Uses in Agriculture

How the network of networks is built and how data finds its way, how the World Wide Web works on top of it, and how farmers, markets and extension services use it in India.

Learning objectives

After this lecture you should be able to:

1. Define the Internet and the World Wide Web and distinguish between them.
2. Outline the history of the Internet and the Web, globally and in India.
3. Describe the components of the Internet: devices, transmission media, service providers, protocols and addresses.
4. Classify networks by size and topology.
5. Explain the TCP/IP model, IP addressing, the Domain Name System and the parts of a URL.
6. Describe how a web page is fetched and list the main Internet services.
7. Discuss uses of the Internet in agriculture, its limitations, and safe-use practices.

Scope: Lecture 6 described databases; most of them are reached through the Internet, which is the subject of this lecture. Smartphone apps for farmers follow in Lecture 14, geospatial portals in Lecture 15, the Internet of Things in Lecture 23, and national digital-agriculture platforms in Lectures 30 and 31.

1. What is the Internet?

A **computer network** is two or more computers linked so that they can share data and resources. The **Internet** is a global **network of networks**: millions of private, public, academic, business and government networks joined together, all exchanging data by a common set of rules — the **TCP/IP protocol suite** (Kurose & Ross, 2021; Leiner et al., 2009). No single organisation owns it; each network is run by its owner, and they agree to carry one another's traffic.

	Internet	World Wide Web (WWW)
What it is	The physical and logical network: cables, radio links, routers and the protocols that move data	A service that runs on the Internet: a huge collection of linked documents and applications
Main protocols	TCP/IP	HTTP/HTTPS, with pages written in HTML and found by URLs
Began	ARPANET, 1969; TCP/IP from 1983	Proposed by Tim Berners-Lee at CERN, 1989
Other services it carries	Email, file transfer, video calls, online games, IoT data — and the Web	—
Analogy	The road network	One kind of traffic on the roads (other traffic: email, video calls)

- **Intranet** — a private network inside one organisation that uses Internet technology (e.g. a university's internal portal for staff and students).

- **Extranet** — part of an intranet opened to selected outsiders (e.g. a seed company's portal for its registered dealers).

2. A short history

Year	Milestone
1969	**ARPANET**, funded by the US Advanced Research Projects Agency, links its first computers; the first host-to-host message is sent from UCLA to the Stanford Research Institute in October 1969 (Leiner et al., 2009)
1974	Vinton Cerf and Robert Kahn publish the design of a protocol for connecting different packet networks — the basis of TCP/IP (Cerf & Kahn, 1974)
1983	ARPANET switches to TCP/IP on 1 January 1983 (Leiner et al., 2009); the Domain Name System is introduced in the same period (Mockapetris, 1987)
1986 (India)	**ERNET** begins as a network for education and research institutions
1989	Tim Berners-Lee at CERN proposes a linked information system (Berners-Lee, 1989) — the start of the **World Wide Web**
1990–91	First web server, browser and website at CERN (Berners-Lee et al., 1994)
1993	On 30 April 1993 CERN makes the Web software freely available to anyone, without royalties (CERN, n.d.)
1995 (India)	Public Internet service opens in August 1995, launched by VSNL in four metro cities
2022 (India)	5G mobile services launched on 1 October 2022
2026 (India)	1,114 million Internet subscriptions at the end of June 2026 (Telecom Regulatory Authority of India [TRAI], 2026)

3. Components of the Internet

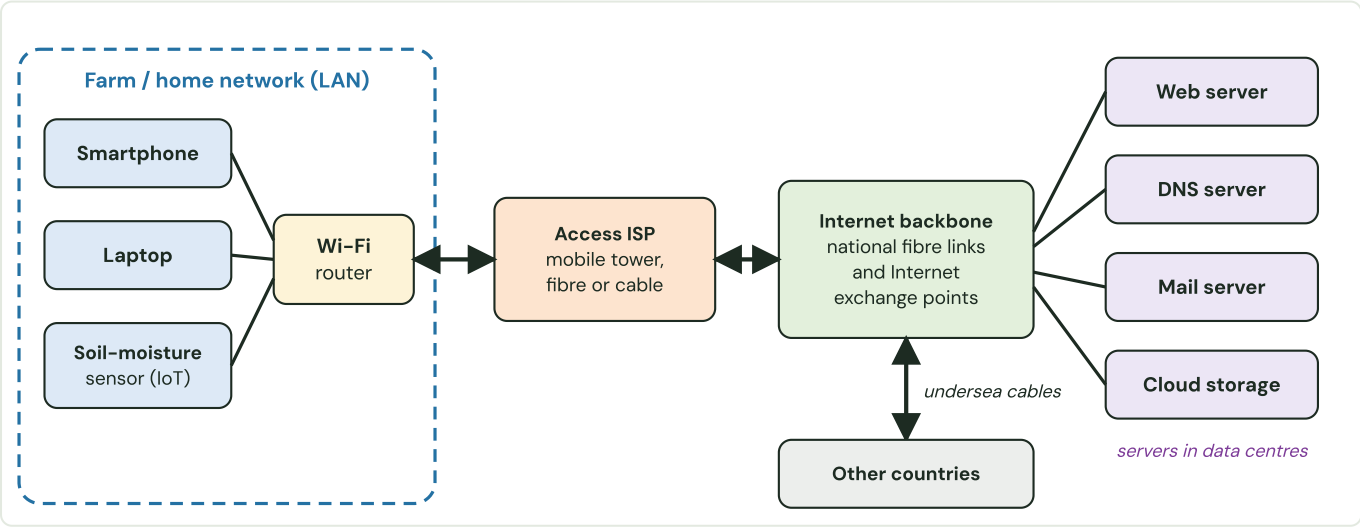


Figure 1. The Internet as a network of networks: a farm network connects through an access provider and the national backbone to servers anywhere in the world. Original diagram.

3.1 Hosts: clients and servers

Every device on the Internet is a **host**. A **client** asks for a service (your phone's browser asking for a price page); a **server** provides it (the computer that stores and sends that page). Most Internet services follow this **client–server** model. In the **peer-to-peer** model, devices share directly with one another without a central server.

3.2 Network devices

Device	What it does
Network interface card (NIC)	Connects a device to a network; each has a unique hardware (**MAC**) address
Modem	**Mo**dulator–**dem**odulator: converts digital data to signals that travel over a telephone line, cable or fibre, and back
Hub	Repeats incoming data to every port; now replaced by switches
Switch	Connects devices within a LAN and sends data only to the device it is meant for (using MAC addresses)
Router	Connects different networks and forwards each packet towards its destination using IP addresses
Wireless access point	Lets Wi-Fi devices join a wired network (home routers usually combine modem, router, switch and access point)
Repeater	Regenerates a weak signal so that it can travel further
Gateway	Joins networks that use different protocols
Firewall	Hardware or software that blocks unwanted traffic according to security rules

3.3 Transmission media

Medium	How it carries data	Typical use
Twisted-pair copper cable	Electrical signals	Ethernet LANs in offices and laboratories; telephone lines
Coaxial cable	Electrical signals, shielded	Cable-TV Internet
Optical fibre	Pulses of light in glass; very high capacity over long distances	Backbones, undersea cables, fibre-to-the-home, village fibre networks
Radio (wireless)	Radio waves	Wi-Fi, Bluetooth, 4G and 5G mobile networks, field sensors
Satellite	Radio links via satellites	Remote areas, ships, disaster sites

3.4 Service providers, protocols and addresses

- **Internet service providers (ISPs)** connect homes and businesses. Access ISPs connect to larger regional and national networks, which interconnect at **Internet exchange points** and reach other countries through **undersea cables**.
- **Protocols** are agreed rules for communication — like a common language and etiquette. The main ones are described in Section 5.
- **Addresses**: every device needs an **IP address** so that data can reach it, and people use easy **domain names** instead (Section 6).

- **Software**: operating-system networking functions, browsers, email programs and apps.

4. Types of networks and topologies

Type	Coverage	Example
PAN — personal area network	A few metres around one person	Phone linked to a smartwatch or wireless earphones by Bluetooth
LAN — local area network	A room, building or campus	Computer laboratory; Wi-Fi in a farmhouse or college
MAN — metropolitan area network	A city	A city-wide cable or fibre network
WAN — wide area network	A country or the world	A bank's national network; the **Internet is the largest WAN**

The **topology** of a network is the arrangement of its devices and links.

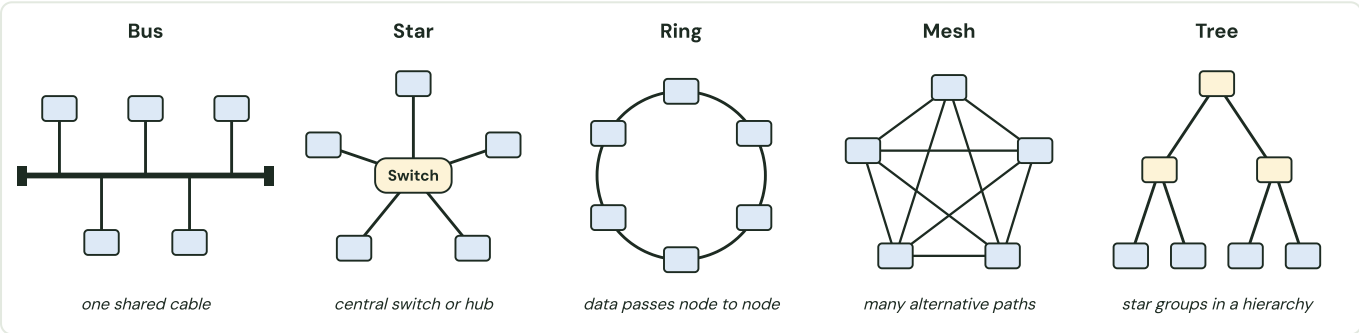


Figure 2. Common network topologies. Original diagram.

Topology	Advantages	Limitations
Bus	Little cable; simple	A break in the main cable stops the whole network; slows as devices are added
Star	Easy to add devices and find faults; one cable failure affects only one device	If the central switch fails, the network stops
Ring	Orderly data flow	One failed node or link can break the ring
Mesh	Very reliable — many alternative paths	Many links; costly. The Internet backbone is a partial mesh
Tree	Expandable; suits large organisations	Failure of a higher-level link cuts off the branch below it

5. How data travels: packets and protocols

The Internet uses **packet switching**: a file or message is cut into small **packets**, each carrying the source and destination addresses. Routers forward each packet independently, and the receiving computer puts them back in order. Lines are shared by many users, so the network is used efficiently, and packets can take another route if a link fails. (A traditional telephone call instead used **circuit switching**, reserving one path for the whole call.)

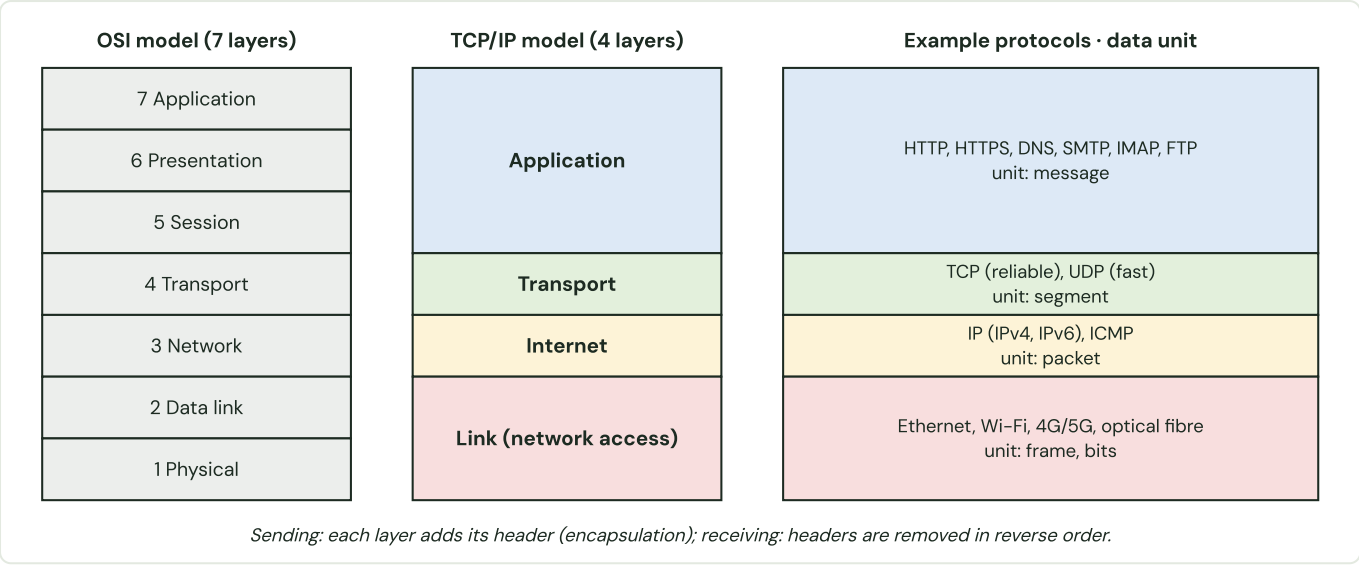


Figure 3. The layered TCP/IP model compared with the seven-layer OSI reference model. Original diagram.

- **Application layer** — the services that users see: web (HTTP/HTTPS), name look-up (DNS), email (SMTP to send; IMAP or POP3 to read), file transfer (FTP).
- **Transport layer** — **TCP** delivers data completely and in order, resending lost packets (web pages, email); **UDP** is faster but does not resend (live video, voice calls).
- **Internet layer** — **IP** gives every packet its source and destination address and routers choose its path.
- **Link layer** — moves data across one physical link: Ethernet cable, Wi-Fi, mobile radio or fibre.
- The **OSI model** (Open Systems Interconnection) is a seven-layer reference model used for teaching and standards; the Internet itself is built on the four-layer TCP/IP model.

6. IP addresses and the Domain Name System

6.1 IP addresses

- **IPv4** addresses are **32 bits** long, written as four numbers from 0 to 255 separated by dots (e.g. 192.168.1.10).
- **IPv6** addresses are **128 bits** long, written in hexadecimal groups (e.g. 2001:db8::1), because the IPv4 supply has run short (Deering & Hinden, 2017).
- Home and office routers give devices **private addresses** (such as 192.168.x.x) and share one public address for the whole network.

> **Worked example — how many addresses?**
> IPv4: 192.168.1.10 = 11000000.10101000.00000001.00001010 in binary (4 × 8 = 32 bits).
> Number of possible IPv4 addresses = 2^{32} = 4,294,967,296 ≈ 4.3 billion — fewer than the number of connected devices today.
> Number of possible IPv6 addresses = 2^{128} ≈ 3.4×10^{38} — enough to give an address to every sensor, tractor and phone many times over.

6.2 Domain names and DNS

People remember names, not numbers. A **domain name** such as market.example.org is read from right to left: **.org** is the **top-level domain (TLD)**, **example** is the domain registered by an organisation, and **market** is a host or subdomain within it. Country-code TLDs include **.in** for India; Indian government

sites use **.gov.in** and academic sites **.ac.in** or **.edu.in**. The **Domain Name System (DNS)** is a worldwide, distributed directory that translates domain names into IP addresses (Mockapetris, 1987).

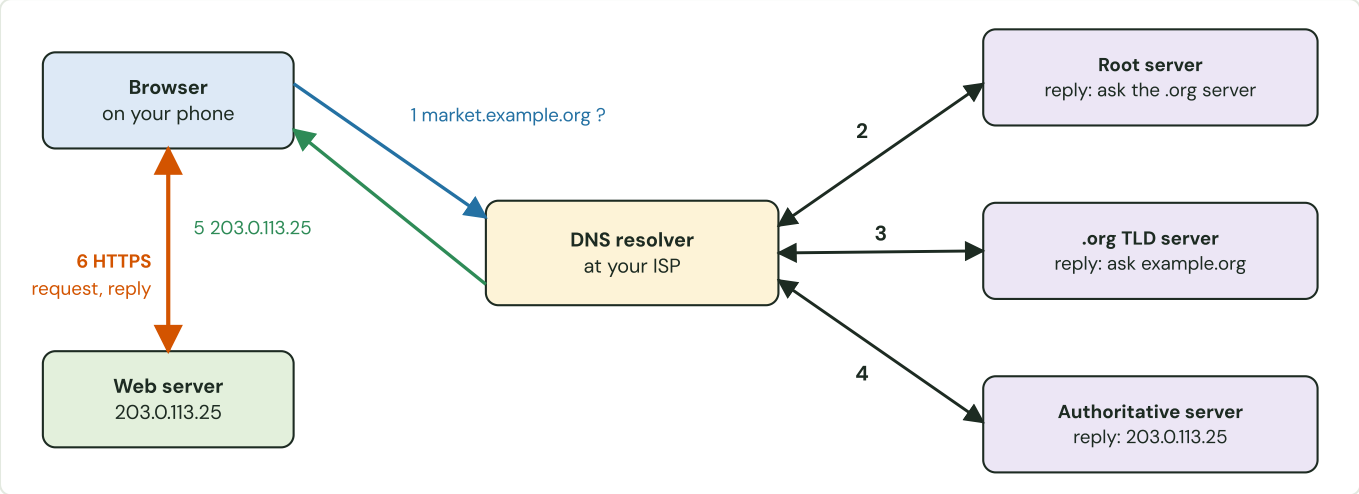


Figure 4. What happens when you open market.example.org: DNS look-up (steps 1–5) followed by the HTTPS request (step 6). Names and addresses are reserved examples. Original diagram.

1. Your browser asks the **DNS resolver** (usually run by your ISP) for the IP address of market.example.org.
2. If the resolver does not already know it, it asks a **root server**, which points to the servers for **.org**.
3. The **.org TLD server** points to the server responsible for **example.org**.
4. The **authoritative server** for example.org returns the IP address.
5. The resolver passes the address back to the browser (and remembers it for a while — **caching**).
6. The browser connects to that address and requests the page with **HTTPS**; the web server sends it back.

7. The World Wide Web

The **World Wide Web** is a system of linked documents and applications that are stored on web servers and viewed in browsers. Berners-Lee's 1989 proposal at CERN described a way to link information across computers with **hypertext** (Berners-Lee, 1989); its three building blocks are still the same today (Berners-Lee et al., 1994):

- **HTML** (HyperText Markup Language) — the language in which web pages are written.
- **URL** (Uniform Resource Locator) — the address of each page or file.
- **HTTP** (HyperText Transfer Protocol) — the rules by which a browser requests a page and a server responds (Fielding et al., 2022). **HTTPS** is HTTP sent over an encrypted connection; the padlock in the address bar shows that the connection is encrypted.

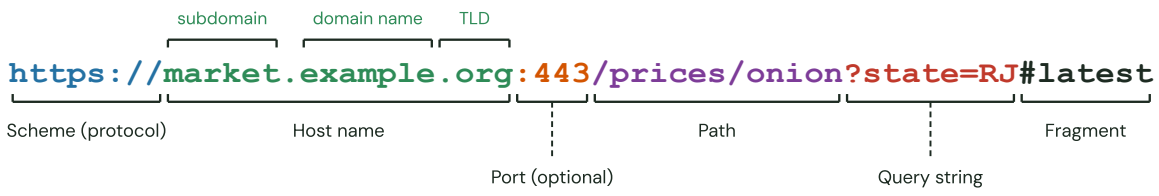


Figure 5. Parts of a URL (example.org is a domain reserved for examples). Original diagram.

Term	Meaning
Web page	A single document on the Web, usually written in HTML
Website	A group of related web pages under one domain
Home page	The main or first page of a website
Hyperlink	Clickable text or image that opens another page or file
Web browser	Client software that requests, displays and links web pages — Chrome, Edge, Firefox, Safari
Web server	Computer and software that store web pages and send them on request
Static vs dynamic page	Static: same content for every visitor. Dynamic: built on request from a database — e.g. today's prices for the market you choose
Web application	A program used through the browser — online forms, e-trading, email in the browser
Search engine	A website that **crawls** the Web, **indexes** pages and **ranks** them for a query — Google, Bing, DuckDuckGo

The early Web (often called **Web 1.0**) was mostly read-only pages. **Web 2.0** describes the interactive Web of user-created content, social media, video sharing and online collaboration. The term **Web 3.0** is used for a more "semantic" (machine-readable) or decentralised Web, but has no single agreed definition. Mobile apps use the same Internet protocols, usually HTTPS, to exchange data with servers.

> **Search tips for agricultural information:** put a phrase in quotes ("drip irrigation subsidy"); add **site:gov.in** or **site:icar.org.in** to search only official sites; add **filetype:pdf** to find bulletins and reports; and always check who published a page and when.

8. Internet services

Service	What it offers	Agricultural example
Email	Messages with attachments to name@domain addresses (SMTP, IMAP, POP3)	Sending a soil-test report to an extension officer
World Wide Web	Information, forms and applications	Checking a scheme's eligibility rules and applying online
Search	Finding information	Locating a pest-management bulletin
File transfer and cloud storage	Uploading, sharing and backing up files	Sharing drone images of a field with a consultant
Video conferencing and VoIP	Voice and video calls over IP	Online training of farmers by a Krishi Vigyan Kendra
Instant messaging and social media	Groups, posts and short videos	An FPO's group sharing meeting dates and price updates
E-commerce and digital payments	Online buying, selling and UPI payments	Ordering seed or paying a machinery-hire charge
E-learning	Courses, recorded lectures and online tests	Online courses in agricultural engineering
Internet of Things (IoT)	Sensors and machines sending data automatically	Soil-moisture sensors switching on a drip system (Lecture 23)

9. Connecting to the Internet

Connection	Medium	Notes
Dial-up	Telephone line	Very slow (tens of kbps); now obsolete
DSL	Telephone copper line	Always on; speed falls with distance from the exchange
Cable	Coaxial cable-TV network	Shared among users in an area
Fibre to the home (FTTH)	Optical fibre	Fastest fixed connection; expanding to towns and villages
Mobile broadband (4G/5G)	Cellular radio	Most Indians reach the Internet this way
Fixed wireless access	Radio link to a fixed receiver, including 5G	Home broadband where fibre has not reached
Satellite	Satellite radio link	Remote and hilly areas; higher delay with geostationary satellites

In India, a connection counts as **broadband** if it can deliver a download speed of at least **2 Mbps** (TRAI, 2026). Speeds are measured in **bits** per second (kbps, Mbps, Gbps), while file sizes are usually given in **bytes** (1 byte = 8 bits).

Worked example — download time and data use (illustrative values)
A 25 MB satellite image of a field = 25 × 8 = 200 megabits.
At 10 Mbps: 200 ÷ 10 = 20 seconds. At the 2 Mbps broadband minimum: 200 ÷ 2 = 100 seconds.
Real downloads take somewhat longer because of protocol overheads and sharing of the network.
A weather app using about 0.5 MB per check, 4 checks a day for 30 days: 0.5 × 4 × 30 = 60 MB a month.

10. The Internet in India

Indicator (end of June 2026)	Value
Total Internet subscriptions	1,114.17 million
Broadband (at least 2 Mbps) / narrowband	1,087.50 million (97.61%) / 26.67 million
Wired / wireless subscriptions	47.73 million / 1,066.43 million
Urban / rural subscriptions	666.09 million / 448.08 million
Subscriptions per 100 population	77.94 overall — urban 129.03, rural 49.07
Average wireless data use	28.39 GB per subscriber per month

Source: TRAI (2026). These are subscriptions, not people: one person may have several SIM cards, which is why the urban figure exceeds 100 per 100 population. About 94% of subscriptions are used through mobile devices.

The gap between urban and rural access — the **digital divide** — matters for agriculture because most farmers live in villages. Government programmes such as **BharatNet**, which lays optical fibre to gram panchayats, aim to close it; these programmes are covered in Lecture 30.

11. Uses of the Internet in agriculture

FAO and ITU describe **e-agriculture** as improving agricultural and rural development through better information and communication, using every kind of ICT — the Internet, mobile phones and sensors as well as radio, television and satellites (Food and Agriculture Organization of the United Nations & International Telecommunication Union, 2016). The Internet carries most of these services today.

Use	Indian examples
Market information and online trade	**e-NAM** (National Agriculture Market), launched in April 2016, links regulated mandis for online trading: 1,656 mandis in 23 states and 4 union territories, with over 1.80 crore farmers and 4,724 FPOs registered and ₹4.84 lakh crore of cumulative trade by March 2026 (Press Information Bureau [PIB], 2026). Daily mandi prices are also published on the AGMARKNET portal (Lecture 14)
Expert advice	**Kisan Call Centre**: launched on 21 January 2004 to answer farmers' queries in their own language on the toll-free number **1800-180-1551** (Department of Agriculture and Farmers Welfare, n.d.); advisories are also sent as SMS through the mKisan portal
Weather and agromet advisories	District forecasts and crop advisories delivered by websites, SMS and apps (Lectures 11 and 13)
Government schemes and records	Online registration and direct benefit transfer; the Soil Health Card portal and AgriStack registries (Lectures 6 and 31)
Learning and extension	University and Krishi Vigyan Kendra websites, online courses, webinars and farmer-to-farmer videos
Research	Online journals, research databases and open data; satellite and map data from geoportals such as ISRO's Bhuvan (Lecture 15)
Inputs, credit and insurance	Online purchase of seed, fertiliser and machinery; UPI payments; online crop-insurance claims
Precision farming and IoT	Sensor, drone and machine data sent to the cloud for analysis and automatic control (Lectures 23–26)

Case study (illustrative) — checking prices before selling
A farmer with 40 quintals of onion compares prices on a market-information portal. Mandi A, nearby, shows ₹1,450 per quintal; Mandi B, 38 km farther, shows ₹1,620 per quintal, and the extra transport costs ₹60 per quintal.
Net gain from selling at Mandi B = (1,620 - 1,450 - 60) × 40 = 110 × 40 = ₹4,400.
Without the Internet the farmer would have had to travel or rely on a trader's word to learn the price difference.

Advantages and limitations

Advantages	Limitations
Timely information on prices, weather, pests and schemes	Weak or no connectivity in many remote areas; rural access lags urban access
Better prices and wider markets; less dependence on middlemen	Cost of smartphones and data for small farmers
Direct access to experts and training	Language, literacy and digital-skill barriers
Transparent, faster government services and payments	Online fraud, fake apps and misinformation
Data for precision farming and research	Power cuts; dependence on the network for critical services

12. Using the Internet safely

Threat	What it looks like	Protection
Phishing	An email, SMS or website that imitates a bank, company or scheme to steal passwords or card details	Check the sender and the exact web address; type official addresses yourself instead of clicking links
Phone (OTP) fraud	A caller claiming to be from a bank or a government office asks for an OTP, PIN or card number	Never share an OTP, PIN or password with anyone
Fake subsidy, loan or prize messages	Promises of money after paying a "fee" or clicking a link	Verify on the official .gov.in site, the Kisan Call Centre or the local agriculture office
Malware and fake apps	Programs that steal data, show adverts or lock files	Install apps only from the official app store and publisher; keep the system updated
Account hacking	Weak or reused passwords are guessed or leaked	Long passphrases, a different password for each account, two-factor authentication
Misinformation	Wrong pesticide doses or false rumours forwarded in groups	Check advice with a KVK, university or the Kisan Call Centre before acting

- **HTTPS and the padlock** show that the connection is encrypted — not that the website is genuine. A fake site can also use HTTPS.
- Avoid banking or entering passwords on open public Wi-Fi.
- Cybercrime, including online financial fraud, can be reported on the national helpline **1930** or at **cybercrime.gov.in** (Ministry of Home Affairs, n.d.).
- Offences such as hacking and identity theft are dealt with under the **Information Technology Act, 2000** (Government of India, 2000); personal data is protected under the DPDP Act, 2023 (Lecture 6).

13. Key takeaways

- The Internet is a global network of networks using TCP/IP; the World Wide Web is one service on it, built on HTML, URLs and HTTP.
- The Internet grew from ARPANET (1969); TCP/IP was adopted in 1983, and the Web was proposed by Tim Berners-Lee at CERN in 1989.
- Its components are hosts (clients and servers), network devices, transmission media, ISPs and backbones, protocols and addresses.
- Networks are classed by size (PAN, LAN, MAN, WAN) and topology (bus, star, ring, mesh, tree).
- Data travels as packets through the four TCP/IP layers; IP addresses (IPv4 32-bit, IPv6 128-bit) identify devices, and DNS translates names to addresses.
- India had 1,114 million Internet subscriptions in June 2026, mostly mobile, but rural access still lags urban access.
- Farmers use the Internet for prices and trade (e-NAM), expert advice (Kisan Call Centre), weather, schemes, learning and precision farming — and need to use it safely.

14. Examination questions

Short-answer questions (2 marks)

Q1. Define the Internet.

A global network of interconnected computer networks that exchange data using the TCP/IP protocol suite.

Q2. Distinguish between the Internet and the WWW.

The Internet is the network itself; the WWW is a service on it — linked pages accessed with browsers through HTTP.

Q3. Expand TCP/IP, HTTP, URL and DNS.

Transmission Control Protocol/Internet Protocol; HyperText Transfer Protocol; Uniform Resource Locator; Domain Name System.

Q4. What is an IP address? Give an example.

A numerical address that identifies a device on a network, e.g. the IPv4 address 192.168.1.10.

Q5. State the function of a router.

It connects different networks and forwards packets towards their destination using IP addresses.

Q6. Name the four layers of the TCP/IP model.

Application, transport, Internet and link (network access).

Q7. What is DNS?

A distributed directory system that translates domain names into IP addresses.

Q8. Who proposed the World Wide Web, and when?

Tim Berners-Lee, at CERN, in March 1989.

Q9. What is the minimum download speed for a broadband connection in India?

2 Mbps.

Q10. Give the toll-free number of the Kisan Call Centre and state its purpose.

1800-180-1551; it answers farmers' queries in their own language.

Long-answer questions (5–10 marks) — answer outlines

Q1. Describe the components of the Internet with a diagram.

Figure 1; hosts (clients, servers); devices (NIC, modem, switch, router, access point, firewall); media (copper, fibre, radio, satellite); ISPs and backbone; protocols; IP addresses and DNS.

Q2. Explain the types of networks and network topologies.

PAN, LAN, MAN, WAN with examples; bus, star, ring, mesh, tree with diagrams (Figure 2), advantages and limitations.

Q3. Explain the TCP/IP model and compare it with the OSI model.

Figure 3; four layers with protocols and data units; packet switching; TCP vs UDP; OSI as a seven-layer reference model.

Q4. Explain how a web page is opened when a URL is typed in a browser.

Parts of the URL (Figure 5); DNS look-up through resolver, root, TLD and authoritative servers; HTTPS request and response (Figure 4).

Q5. Discuss the uses of the Internet in agriculture with Indian examples.

e-NAM, AGMARKNET, Kisan Call Centre and mKisan, weather advisories, schemes and records, learning, research, inputs and payments, IoT; case study; advantages and limitations; digital divide.

Q6. What are the common online threats faced by farmers, and how can they be avoided?

Phishing, OTP fraud, fake schemes, malware, hacking, misinformation; protections; HTTPS; reporting on 1930 and cybercrime.gov.in; IT Act, 2000.

15. Glossary

Term	Meaning
Bandwidth	The data-carrying capacity of a connection, in bits per second
Browser	Software for viewing web pages
Client	A device or program that requests a service
DNS	Domain Name System — translates domain names into IP addresses
Domain name	The human-readable name of a host or website, e.g. example.org
Firewall	A barrier that filters network traffic according to security rules
HTML	HyperText Markup Language, used to write web pages
HTTP / HTTPS	Protocol for requesting and sending web pages; HTTPS is the encrypted version
Hyperlink	A link from one page or file to another
Intranet	A private network within an organisation using Internet technology
IP address	Numerical address of a device on a network (IPv4: 32 bits; IPv6: 128 bits)
ISP	Internet service provider
LAN / WAN	Local area network / wide area network
Modem	Converts digital data to line signals and back
Packet	A small unit of data sent independently across a network
Protocol	An agreed set of rules for communication
Router	A device that forwards packets between networks
Server	A computer or program that provides a service to clients
TLD	Top-level domain, such as .in, .org or .com
URL	Uniform Resource Locator — the address of a resource on the Web

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All diagrams are original. Explanations are written in original language; standard content on networks, protocols, addressing and the Web follows Kurose and Ross (2021) and the cited RFCs. Names such as market.example.org and addresses such as 203.0.113.25 are reserved for examples. Prices in the case study are illustrative.

Practice MCQs

Q1. The Internet grew out of which network?

(a) ERNET (b) ARPANET
(c) NICNET (d) BharatNet

Q2. The World Wide Web was proposed by

(a) Vinton Cerf (b) Tim Berners-Lee
(c) E. F. Codd (d) Charles Babbage

Q3. Which protocol is used to request and send web pages?

(a) SMTP (b) FTP
(c) HTTP (d) UDP

Q4. DNS translates

(a) IP addresses into MAC addresses (b) domain names into IP addresses
(c) HTML into text (d) analogue signals into digital

Q5. An IPv4 address is how many bits long?

(a) 16 (b) 32
(c) 64 (d) 128

Q6. Which device forwards packets between different networks?

(a) Hub (b) Repeater
(c) Router (d) Modem

Q7. In which topology are all devices connected to a central switch?

(a) Bus (b) Ring
(c) Star (d) Mesh

Q8. IP works at which layer of the TCP/IP model?

(a) Application (b) Transport
(c) Internet (d) Link

Q9. In https://market.example.org/prices, "market.example.org" is the

(a) scheme (b) host (domain) name
(c) path (d) query string

Q10. Which protocol is used to send email?

(a) SMTP (b) HTTP
(c) DNS (d) IMAP

Q11. In India, the minimum download speed for a broadband connection is

(a) 256 kbps (b) 512 kbps
(c) 2 Mbps (d) 10 Mbps

Q12. The toll-free number of the Kisan Call Centre is

(a) 1930 (b) 112
(c) 1800-180-1551 (d) 100

Answers

Q1 — (b) ARPANET
Q2 — (b) Tim Berners-Lee
Q3 — (c) HTTP
Q4 — (b) domain names into IP addresses
Q5 — (b) 32
Q6 — (c) Router
Q7 — (c) Star
Q8 — (c) Internet
Q9 — (b) host (domain) name
Q10 — (a) SMTP
Q11 — (c) 2 Mbps
Q12 — (c) 1800-180-1551